

Furniture Product Launch Advisor

Predicting consumer interest in IKEA furniture products

398

IKEA products

10

product categories

Live

FastAPI API

Live

Streamlit app

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App: <https://ikea-advisor.streamlit.app/>

API: <https://ikea-product-interest-api.onrender.com>

Project Overview & Motivation

From product listings to launch decision support

Problem

Furniture retailers manage large catalogs, but it is hard to know which products will attract customer attention.

Product teams need early signals before investing in launches, pricing, inventory, and marketing.

Why It Matters

Poorly positioned products can fail to gain traction even if they are well designed.

A data-driven interest estimate can reduce launch risk and support pricing decisions.

Key Objectives

Collect real IKEA product data from category and product pages.

Define consumer interest using review count as a proxy for engagement.

Build and deploy a model-powered product launch advisor.

Scrape
products

Build
features

Predict
interest

Advise
launch

Data Collection & Preprocessing

Final dataset characteristics and cleaning pipeline

398

products

10

categories

168

median reviews

4.32

avg rating

Collection

Source: IKEA US product pages

Category pages produced product URLs

Detail pages provided price, rating, review_count

Preprocessing

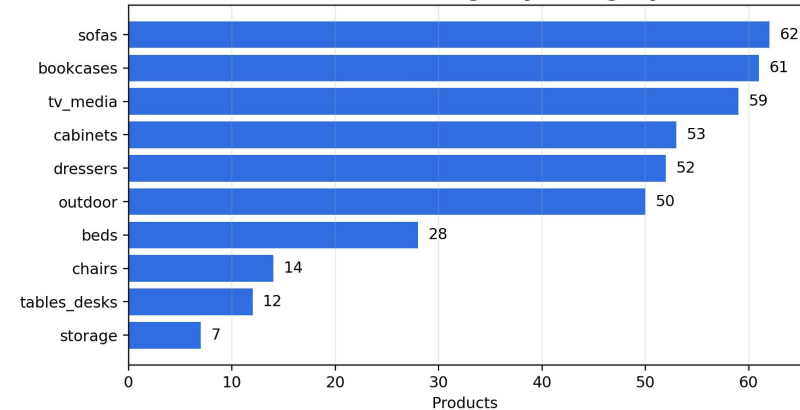
Deduplicated by product_url

Converted price/rating/reviews to numeric fields

Standardized category labels

Target: high_interest = review_count above median

Product Coverage by Category



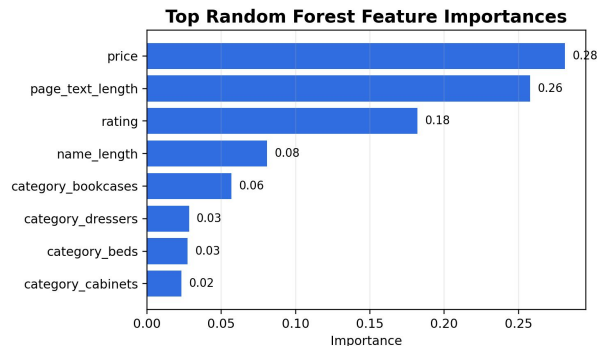
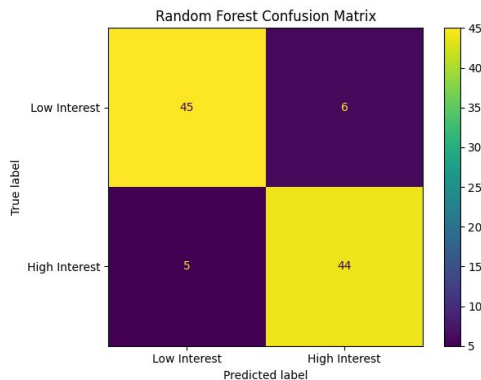
Main challenge: consumer interest is not directly observable, so review_count is used as an engagement proxy. Category pages did not expose these signals, so detail pages were scraped.

Model Development & Evaluation

All metrics reported on held-out test set (100 products)

Feature Set

category (one-hot encoded)
price
rating
name_length
page_text_length



Interpretability Insights

Price was the strongest predictor of high review engagement.

Product page length may serve as a proxy for product maturity and information richness. Products with more detailed pages tended to accumulate higher customer engagement.

Rating added additional signal beyond price and category.

Exceeded success criterion ($F1 \geq 0.70$) — RF $F1 = 0.889$.

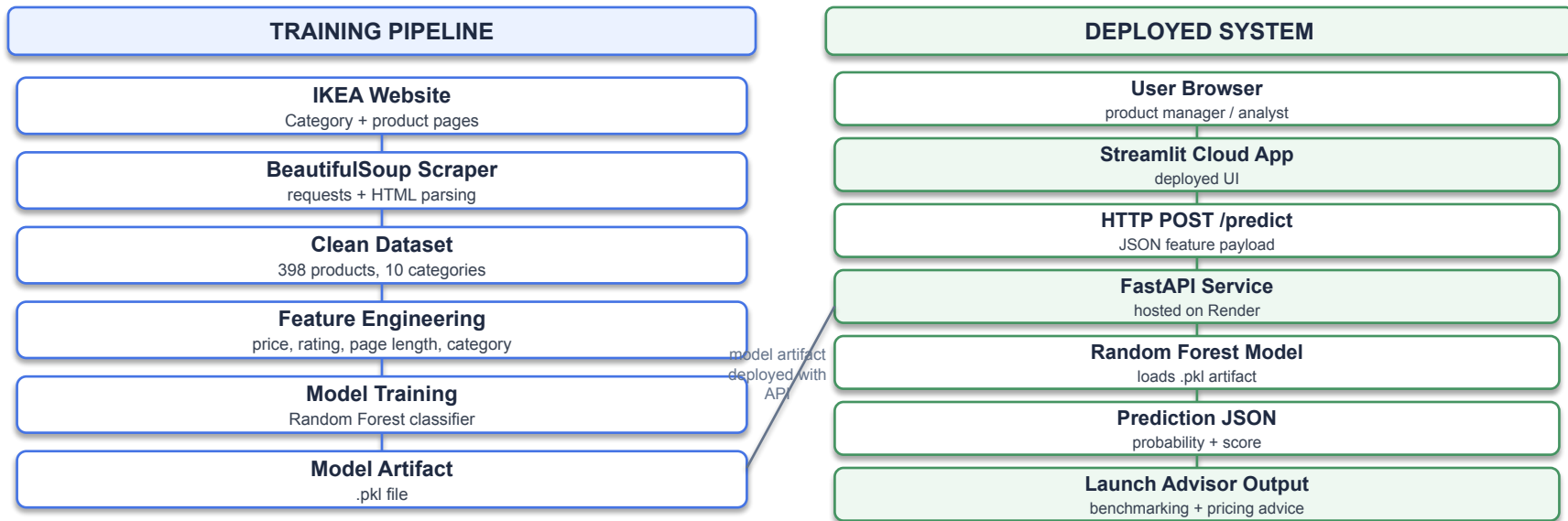
Model	Accuracy	Precision	Recall	F1
Logistic Regression	0.82	0.804	0.84	0.82
Random Forest	0.89	0.88	0.90	0.889

Test set: 100 products • Correct: 89 • FP: 6 • FN: 5

Solution Architecture

05

End-to-end deployed system: training pipeline + cloud app/API communication



App: ikea-advisor.streamlit.app • API: ikea-product-interest-api.onrender.com

Application Features & Business Use Cases

06

Deployed Streamlit app for launch planning, benchmarking, and pricing analysis

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Furniture Product Launch Advisor

Product Inputs

Product category

beds

Planned price (\$)

159.00

Expected customer rating

4.50

Product information level

Rich

Furniture Product Launch Advisor

This tool helps product teams estimate whether a furniture product is likely to receive high consumer interest. It uses IKEA product data including category, price, rating, and product page information.

Prediction Result

Likely High Interest

Predicted Interest Score

77/100

This product has strong launch potential based on the model.

Category Benchmark

Category Avg Price	Category Avg Ra...	Median Reviews
\$334	4.07	389

Business Recommendation

- Price is below the category average, which may support stronger customer interest.
- Expected rating is above the category average, which supports customer trust.
- Product information is relatively rich, which may help customers evaluate the product.

Recommended Action: Launch with current positioning.

Pricing What-if Analysis

	Scenario	Price	Interest Score
0	10% Lower Price		143.1
1	Current Price		159
2	10% Higher Price		174.9

Best scenario in this test: 10% Lower Price with an interest score of 80/100.

Price Sensitivity Analysis

Core Use Cases

1. Launch Risk

Predict high vs. low interest and convert probability into a 0–100 Interest Score.

2. Benchmarking

Compare planned price and rating with category baselines and median reviews.

3. Pricing What-if

Test alternative prices and identify better launch-price scenarios.

Example output

Bookcase, \$60, rating 5.0 → Score 86/100; recommended action: launch.

Interest Score = predicted probability of high-interest class × 100

Challenges, Learnings & Future Work

What changed during development and where the project can go next

Technical Challenges

Category pages did not expose full product popularity data.

Initial target label was derived from `rank_in_category`, which was also included as a feature — causing target leakage.

Resolved by redefining the target using `review_count` and removing leakage-related features.

Product variants created near-duplicate records.

Final solution scraped detail pages and used `review_count` as the interest proxy.

Key Learnings

Target definition is as important as model selection.

A leakage issue in the initial label inflated accuracy. Correcting the target created a more realistic evaluation.

Feature engineering contributed more than algorithm choice.

Adding category, price, rating, and page-level features improved F1 from ~0.50 to 0.89.

Business value comes from decision support, not prediction alone.

Transforming the model into a launch advisor with pricing analysis made the system more actionable.

Future Work

Expand beyond IKEA to more furniture brands.

Add image features from product photos.

Predict continuous `review_count` instead of a binary label.

Build a stronger pricing optimization module.

Role of AI assistants (ChatGPT/Cursor): helped identify target leakage, debug scraping failures, generate visualization code, and improve the business framing of the launch advisor; final model evaluation and interpretation were verified manually.